Project IV Report

Organizational Performance Analysis Using Process Mining and Data-Driven Weighting

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Index

1 Title Page……..1

2 Candidates Declaration and Supervisors Declaration…….3

3 Acknowledgement……4

4 Abstract…….5

5 Introduction……..6

6 Literature Review…….7

7 Dataset Overview…….8

8 Methodology……10

9 Result……...…...13

10 Conclusion…….22

11 References……..24

CANDIDATE'S DECLARATION

We hereby declare that the work presented in this report titled *"Data-Driven Workplace Performance Measurement Using Process Mining and Machine Learning"* submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in the School of Engineering and Technology, BML Munjal University, is an original work carried out by the undersigned under the guidance of Dr. Atul Mishra. The work has not been submitted elsewhere for any other degree or diploma.

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Abhinav Vuddagiri

SUPERVISOR'S CERTIFICATE

This is to certify that the above statement made by the candidates is true to the best of my knowledge and that the project has been carried out under my supervision.

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Dr. Atul Mishra has provided great help in carrying out our work and is acknowledged with reverential thanks. Without wise counsel and able guidance, it would have been impossible to complete the project in this manner.

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ABSTRACT

This study presents a comprehensive, data-driven framework for workplace performance measurement using the Event Graph representation of the BPIC 2015 municipal permit process. The original GraphML dataset, which models events, entities, and behavioral relationships in a labeled property graph, was reconstructed into a graph-preserving CSV structure to enable multi-level analytical processing while retaining essential semantics such as event order, entity correlations, and resource interactions. Using this enriched dataset, we developed a full performance evaluation pipeline consisting of multi-level feature engineering, hybrid outlier detection, PCA–entropy–based data-driven weighting, and integrated scoring models for cases, resources, and municipalities.

Results show that the municipal permit process exhibits strong behavioral heterogeneity across cases, substantial workload imbalance across human resources, and clear structural differences across municipalities. Data-driven weighting schemes assign higher importance to features that explain the greatest behavioral variance particularly outlier penalties, event volume, handover complexity, and structural indicators resulting in more discriminative and meaningful performance scores than standard manual weighting. Outlier-aware hybrid scoring amplifies distinctions between typical and irregular behaviors, improving interpretability and ranking accuracy. Variant-level analysis reveals that the most frequent execution paths are not necessarily the best-performing, highlighting targeted opportunities for process improvement.

The resulting dashboards provide actionable diagnostic insights by visually summarizing complex performance patterns across organizational levels, enabling managers to identify bottlenecks, anomalous behaviors, and structural inefficiencies. By integrating graph-derived semantics with data-driven scoring and multi-level analytics, this project extends the capabilities of traditional process mining and bridges conceptual gaps in the Workplace Performance Measurement (WPM) and event-graph literature. The proposed framework demonstrates a scalable, behaviorally grounded approach to high-resolution performance analysis, offering a practical foundation for optimization and data-driven decision-making in real-world organizational settings.

Keywords - Workplace Performance Measurement, Process Mining, Event Graphs, GraphML Reconstruction, Behavioral Analytics, Multilevel Performance Scoring, Data-Driven Weighting, Outlier Detection, Case Analysis, Resource Analysis, Municipality Performance, Business Process Intelligence

Introduction

Workplace performance measurement has evolved over more than a century, beginning with classical theories of scientific management that emphasized task decomposition, time measurement, and labor optimization. Taylor's foundational work on scientific management established systematic performance observation as an essential component of organizational productivity improvement]. The motion-study techniques developed by the Gilbreths extended these principles through detailed analysis of physical task execution to reduce waste and improve efficiency. Subsequent advances in statistical quality control, particularly Tippett's early applications of statistical methods to industrial processes , introduced quantitative rigor to performance analysis, enabling organizations to understand process variability and detect inefficiencies. Modern workplace performance research further matured with the development of work sampling approaches, which enabled large-scale behavioral observations across complex environment However, these traditional techniques were largely limited to manual measurement and lacked the ability to capture the full behavioral richness of contemporary digital processes. The emergence of event data and process mining introduced a transformative shift, enabling automated, data-driven performance evaluation through the extraction of behavioral patterns from event logs. Recent business process intelligence initiatives, including the BPI Challenges, have provided large-scale, real-world datasets such as BPIC 2015, enabling deeper analysis of organizational processes in public-sector settings.

Despite these developments, traditional event logs lack the structural richness required to fully represent complex, multi-entity behavior. Event Graph representations address this limitation by modeling processes in labeled property graphs that capture multi-dimensional relations between events, entities, and behavioral constraints. Esser and Fahland formalized this paradigm through their multi-dimensional event graph model, enabling richer queries and analytical methods than classical tabular logs. Parallel work in object-centric process mining (OC-PM) and graph-based process mining further emphasized the need for multi-entity representations that reflect the true relational structure of modern processes. Building upon this theoretical foundation, recent studies on workplace performance digitalization demonstrate the potential of data-driven behavioral monitoring to provide high-resolution insights into worker performance and organizational efficiency.. However, there remains a significant methodological gap in unifying event-graph-based process representations with multi-level (case, resource, municipality) performance scoring frameworks. This project directly addresses that gap by reconstructing the Event Graph BPIC 2015 dataset into a graph-preserving analytical dataset and developing a multi-layer data-driven performance measurement architecture.

Literature Review

Early foundations of workplace performance measurement derive from classical industrial engineering. Taylor's principles of scientific management emphasized standardized measurement of work tasks using systematic observation and timing [1], while the Gilbreths expanded the methodology through motion-study analysis aimed at minimizing unnecessary operator movement [2]. These foundational studies established the earliest quantitative methods for understanding human work behavior. Tippett advanced this tradition by formalizing the use of statistical methods to quantify process variability and detect inefficiencies [3], while Robinson's work on work sampling provided more scalable frameworks for measuring activity patterns and workforce utilization across large organizational settings [4].

The emergence of process mining reshaped performance analytics by enabling automated extraction of behavioral models from event logs. van der Aalst's seminal work established the theoretical foundations of process mining, including process discovery, conformance checking, and performance analysis [5], unlocking the potential for systematic behavioral evaluation at scale. The BPI Challenge series contributed significantly to empirical research by publishing real-world public-sector event logs, including the BPIC 2015 building permit dataset used in this project [6].

Traditional event logs, however, treat each event as belonging to a single "case," limiting their ability to represent multi-entity interactions. Event Graph representations address this limitation through graph-structured models in which events are associated with multiple entities and semantic relationship types. Esser and Fahland's Event Graph model [7] provides an expressive formalism capturing directly-follows constraints, entity correlations, and class-level abstractions within a labeled property graph. OC-PM, introduced by Berti and van der Aalst [8], extends this paradigm through object-centric logs that generalize traditional case-based event modeling. Jalali's work on graph-based process mining [9] further demonstrates the analytical advantages of graph structures for modeling complex, multi-object processes.

Contemporary research has also investigated digitalization of workplace observation. Nesterak et al. [10] highlight the advantages of sensor-driven behavioral monitoring for performance measurement, demonstrating the growing relevance of data-driven techniques for organizational diagnostics. Across these strands of literature, a clear gap emerges: while multi-dimensional process representations exist, and while performance measurement frameworks exist, the integration of graph-preserving process reconstruction with hierarchical performance scoring remains underexplored. This project contributes directly to closing that gap.

Dataset Overview

The dataset used in this project originates from the BPI Challenge 2015, which provides five separate XES-format event logs, each corresponding to the building permit process of one Dutch municipality. These logs contain detailed case-level information over a four-year period, including activity codes, Dutch and English labels, timestamps, resources, costs (SUMleges), and subprocess indicators. Although the five municipalities are expected to follow the same procedural blueprint, historical changes in regulations, organizational structures, and local workflows introduce natural variation across the logs. The raw XES data therefore provides a rich foundation for analyzing differences in control flow, throughput time, resource behavior, and cross-municipality performance. However, because the logs are separate, the original dataset does not unify entities (e.g., resources working in multiple municipalities) nor provide an integrated view of activities, handovers, and structural relationships across organizational boundaries.

To address these limitations, the Event Graph BPIC 2015 reinterpretation transforms all five XES logs into a single labeled property graph, represented as a Neo4j dump and a `.graphml` file, where events, entities, classes, and logs become nodes connected through structured relationships such as `:DF` (directly-follows), `:CORR` (event–entity correlation), and :OBSERVES (event–class linkage). This graph representation retains the multi-dimensional nature of the underlying processes and unifies the municipalities into one integrated event space, enabling richer structural and behavioral analysis. In our project, this GraphML file served as the primary data source. Since `.graphml` cannot be directly used in traditional process analytics pipelines, we reconstructed a CSV-based event log by exporting only event nodes and their essential properties while preserving graph semantics through carefully rederived relationships. Using attributes from the graph (e.g., correlated entity IDs, timestamps, activity classes, and municipality membership), we rebuilt the case–event linkage, event ordering, resource assignments, and cross-entity relationships needed for reliable analysis. This approach ensured that even though the dataset was converted to tabular form, the behavioral and relational richness of the original graph structure was maintained, forming a robust analytical basis for our multi-level performance feature extraction, outlier detection, and hybrid scoring system.

In comparison to other event graph datasets, the Event Graph BPIC 2015 collection represents an exceptionally strong use case for this project due to its combination of scale, diversity, structure, and real-world process complexity. Unlike synthetic event graph datasets often small, artificially balanced, or lacking multi-entity correlations the BPIC-2015 graph integrates five independent event logs into a unified, highly connected property graph with over 260,000 nodes and more than 2.6 million relationships. It includes multiple resource types, cross-municipality entity sharing, rich subprocess structure, multilingual labels, and cost-related metadata, offering a uniquely multi-dimensional foundation for performance analysis. Importantly, the integrated graph retains both control-flow and organizational perspectives, making it ideal for case-level, resource-level, and municipality-level scoring. Other event graph datasets rarely combine all these dimensions into one coherent structure. By starting from this graph, and then converting it into a structured CSV while preserving graph-based semantics, our project gains a rare opportunity to perform WPM-inspired, multi-layered performance measurement on a dataset that is both rich enough for machine learning and realistic enough to reflect true organizational behavior.

GraphML Component	Description in Event Graph	Reconstructed CSV Field	How It Is Derived / Preserved
:Event node	Represents a discrete process event with timestamp & activity	timestamp, activity, caseid, resource, municipality	Extracted directly from event node attributes; case resource inferred using:CORR relationships
:Entity node	Represents objects and actors (cases, resources, municipalities)	caseid, resource, org:group	Mapped reading entity types connecting correlated entities events
:Class node	Represents activity class/category for events	concept:name, taskNameNL, taskNameEN	Activity metadata attached to event via :OBSERVES relationship
:CORR (Event → Entity)	Links events to cases, resources, and other entities	caseid, resource, application_type	Used to reconstruct case structure and resource assignments
:DF (Directly-Follows)	Defines ordering of events within each entity context	event_order (implicit), chronological ordering	Recreated by sorting all events for each case by timestamp
:HAS (Log → Event)	Groups events belonging to same municipality log	municipality	Municipality ID preserved as organizational grouping
:REL (Entity ↔ Entity)	Represents structural links between entities (e.g., case → responsible actor	*Not directly stored*, used to enrich interpretation	Helps validate entity–event relationships; preserved indirectly through correlation

Table 1. Mapping of GraphML structural components to reconstructed CSV fields.

The table summarizes how core elements of the Event Graph BPIC 2015 dataset such as event nodes, entity nodes, correlation links, and directly-follows relationships were transformed into tabular fields. This mapping illustrates how graph semantics were preserved during the GraphML-to-CSV reconstruction to ensure that case flows, resource associations, and organizational context remained intact for reliable downstream analysis.

Methodology

Figure 1. Workflow pipeline of the performance analysis system

This diagram illustrates the complete end-to-end workflow of the project, beginning with the extraction and transformation of the Event Graph BPIC 2015 dataset and progressing through data cleaning, hierarchical feature engineering, hybrid outlier detection, multi-level scoring, and the final integration into a searchable performance analytics interface.

The pipeline preserves graph-based semantics throughout the transformation from GraphML to CSV, ensuring that relationships between cases, events, resources, and municipalities remain intact for reliable analysis.

The methodological foundation of this project begins with transforming the Event Graph BPIC 2015 dataset from its graph-structured representation into an analysis-ready tabular format. The original GraphML file encodes events, entities, and their relationships such as directly-followed (DF), event–entity correlations (CORR), and class associations using a labeled property graph structure. To ensure analytical compatibility while preserving the inherent relational semantics, only well-defined event nodes and their essential attributes were extracted. Each event retained its timestamp, activity identifier, resource linkage, and municipality membership. Graph relationships were reinterpreted algorithmically by ordering events chronologically within their correlated case entities and reconstructing implicit case flows. This approach maintained the behavioral structure of the graph, producing a CSV event log in which event order, entity interaction, and organizational context remained faithful to the original multidimensional graph model. A series of normalization steps established a consistent schema for downstream analysis, unifying attribute formats and resolving inter-municipality entity overlaps.

Following reconstruction, the data underwent a comprehensive cleaning and standardization process designed to ensure internal consistency and remove structurally invalid entries. Rows with missing or corrupted core attributes such as timestamps, activity labels, or case identifiers were removed to eliminate analytical noise. The timestamp format was harmonized, and syntactic variations in activity nomenclature were normalized across the five municipalities. Duplicate entries and inconsistent event sequences were corrected by applying temporal ordering constraints derived from the graph's original DF semantics. Contextual attributes such as cost, case status, and resource identifiers were validated for numeric and categorical coherence. This produced a final, clean event log in which each record represented a precise and reliable behavioral unit suitable for multi-level analytics.

Feature engineering was conducted across three hierarchical layers case, resource, and municipality to capture diverse behavioral perspectives of the process. At the case level, features such as cycle time, active processing time, waiting times, event counts, subprocess coverage, structural complexity, and cost attributes were derived. These metrics characterize how a permit application evolves, how much work is performed on it, and how efficiently it progresses through the workflow. At the resource level, features emphasized work patterns and collaboration, including workload distribution, event handling diversity, resource handovers, incoming and outgoing interactions, temporal availability, and cross-municipality involvement.

These metrics reveal the operational behavior, specialization, and coordination patterns of human actors across the organization. Municipality-level features aggregated case and resource behaviors into broader structural indicators: workload imbalances, cross-actor coordination networks, handover density, subprocess performance consistency, and normalized performance indices. By combining fine-grained event behaviors into hierarchical performance descriptors, the feature engineering stage established a multi-layered view of the process that supports nuanced and comparative assessment.

To ensure analytical robustness, a hybrid outlier detection strategy was applied to all feature sets. Statistical methods such as interquartile range (IQR), Z-scores, and median absolute deviation (MAD) captured distribution-based anomalies, particularly in skewed metrics like waiting times and case complexity. For more complex, multidimensional deviations, an Isolation Forest model detected outliers that arise from non-linear interactions among features. This hybrid approach ensured that anomalies were not only statistically defined but behaviorally meaningful, reducing the influence of irregular cases or resource behaviors that could distort performance scoring. Outlier flags were incorporated into the feature datasets, enabling the scoring system to incorporate both the existence and severity of anomalous behaviors.

The performance scoring methodology integrates all feature dimensions using a weighted hybrid scoring architecture. Each feature set underwent normalization to ensure comparability, followed by dimensionality reduction through principal component analysis (PCA) to identify variance patterns and reduce redundancy. Feature weights were computed through a combination of entropy-based weighting and PCA loadings, balancing the information contribution of each feature with its behavioral significance. These weighted metrics were aggregated into composite performance scores at the case, resource, and municipality levels. Case scores evaluated processing efficiency and behavioral consistency; resource scores assessed competence, productivity, and collaboration quality; and municipality scores summarized structural and organizational performance. To maintain interpretability and comparability, all scores were normalized onto a unified scale, resulting in performance indices suitable for ranking, benchmarking, and cross-layer analysis.

Finally, the project culminated in integrating these scores into a unified analytical environment. Performance indices were consolidated into reference tables that joined case, resource, and municipality performance in a coherent analytical structure. A lightweight search and exploration interface enabled users to query cases, resources, or municipalities based on their performance metrics, outlier indicators, or structural characteristics. This environment supports targeted process inspection, anomaly investigation, and comparative performance analysis across organizational units. By combining graph-preserving reconstruction, multi-level feature engineering, robust anomaly detection, and hybrid scoring, this methodology provides a complete pipeline for data-driven workplace performance measurement aligned with WPM-inspired principles.

Results

Case-Level Results

The case-level analysis produced a highly detailed behavioral assessment of process instances. The case performance dashboard **(Figure 2)** visualizes eight critical indicators cycle time, active processing time, waiting time, variant length, rework count, total cost, event count, and hybrid score sorted to reveal distributional patterns. These continuous metric curves demonstrate substantial right-skew across multiple attributes, indicating that while many cases progress smoothly, a small but impactful set of cases exhibit extreme delays, large variants, or high costs. This heterogeneity, made visible through the dashboard, confirms the presence of structurally complex or irregular case trajectories that contribute disproportionately to overall process inefficiency.

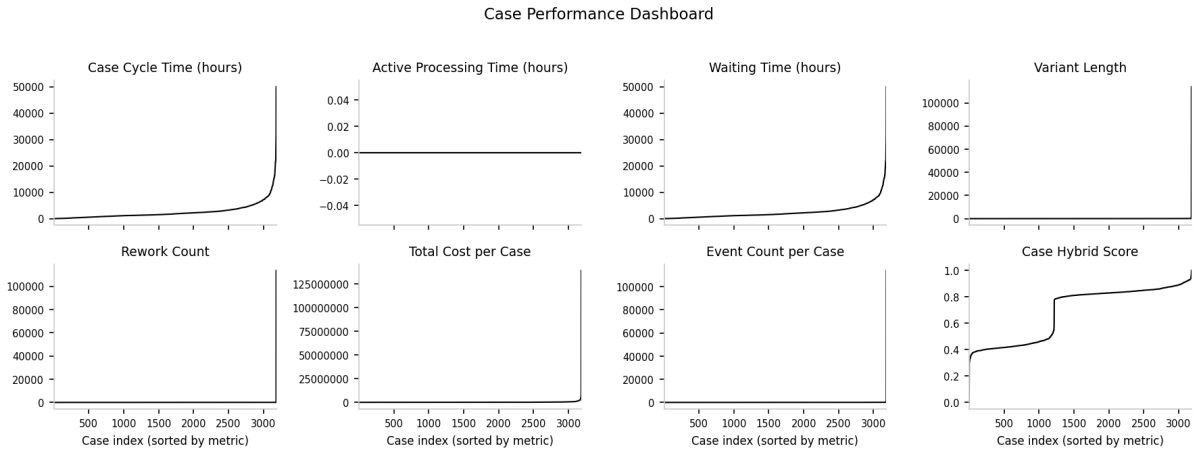

Figure 2. Case-level performance dashboard

This figure presents eight key case-level behavioral metrics sorted and plotted across the entire dataset. Metrics include cycle time, active processing time, waiting time, variant length, rework count, total cost, event count, and the final hybrid case score. The strong right-skew across several metrics highlights structural heterogeneity and the presence of a small number of highly complex or delayed cases.

A central contribution of the scoring framework is the integration of anomaly detection into hybrid scoring. The influence of this penalty mechanism is shown in **Figure 3**, comparing hybrid case scores with and without the outlier penalty. Without penalty, the distribution is overly narrow, concealing distinctions between typical and anomalous cases. After applying the penalty, the distribution separates clearly, pulling irregular cases toward the low-score region. This demonstrates that the hybrid scoring system becomes substantially more discriminative when anomalies are incorporated, preventing irregular cases from appearing artificially average.

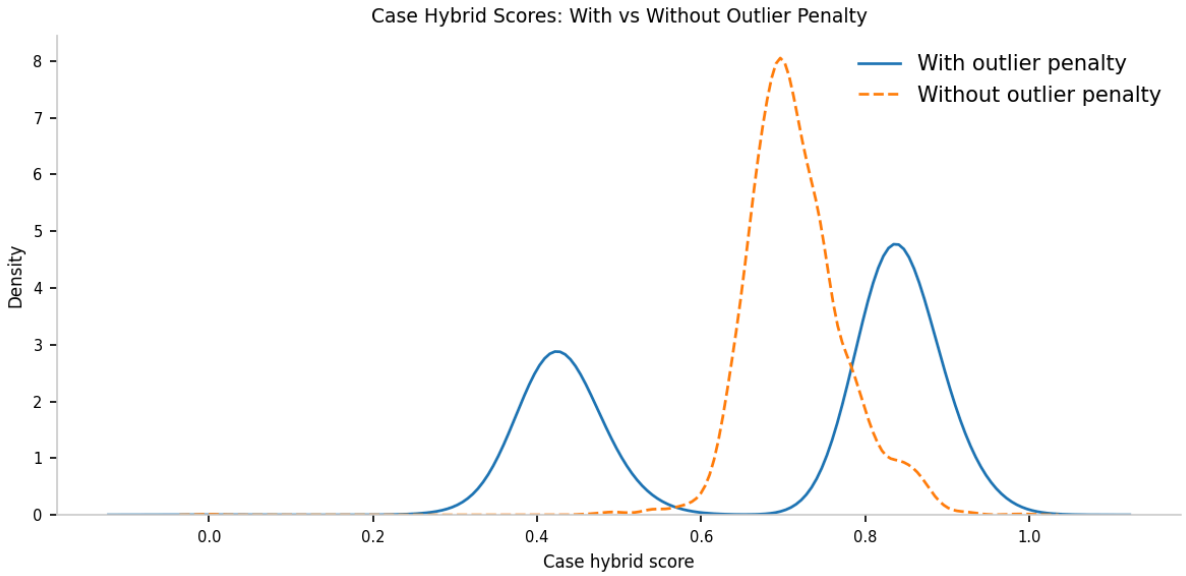

Figure 3. Distribution of case hybrid scores with and without outlier penalties

Kernel density curves show how the outlier penalty separates anomalous cases from the main population. The penalized distribution exhibits greater spread and clearer differentiation between high- and low-performing cases.

To further contextualize performance, the analysis examined process variants. **Table 2** lists the five most frequent variants, along with their average hybrid scores. Although these variants occur most often, their hybrid scores fall in a mid-range, indicating that frequent behaviors are not necessarily optimal. This highlights a key process improvement opportunity: optimizing high-frequency variants yields the largest performance gains system-wide, even if they are not the worst-performing.

Top 5 variants with average hybrid score:

variant	count	avg_hybrid_score
register submission date request->enter senddate acknowledgement->phase application received->reception through OLO->send confirmation receipt	12	0.466333
register submission date request->OLO messaging active->applicant is stakeholder->phase application received->send confirmation receipt->send confirmation receipt->enter senddate acknowledgement	11	0.464364
register submission date request->OLO messaging active->send confirmation receipt->phase application received->send confirmation receipt->enter senddate acknowledgement	8	0.485125
register submission date request->phase application received->reception through OLO	5	0.466600
register submission date request->enter senddate acknowledgement->phase application received->reception through OLO->send confirmation receipt->forward to the competent authority->regular procedure without MER	4	0.486750

Table 2. Top five most frequent process variants and their hybrid performance

This table lists the five most common execution paths by frequency, along with their average hybrid scores. The results show that frequent paths do not always correspond to high performance, indicating optimization opportunities in high-volume process variants.

The effect of data-driven weights at the case level is displayed in **Figure 4**, which compares manual standard weights to the PCA+entropy-derived weights.

The data-driven scheme increases the importance of features that most strongly explain behavioral variance, particularly the outlier penalty and resource-related metrics.

This adjustment materially influences final case rankings, demonstrating that traditional manual weighting understates the performance implications of behavioral irregularity.

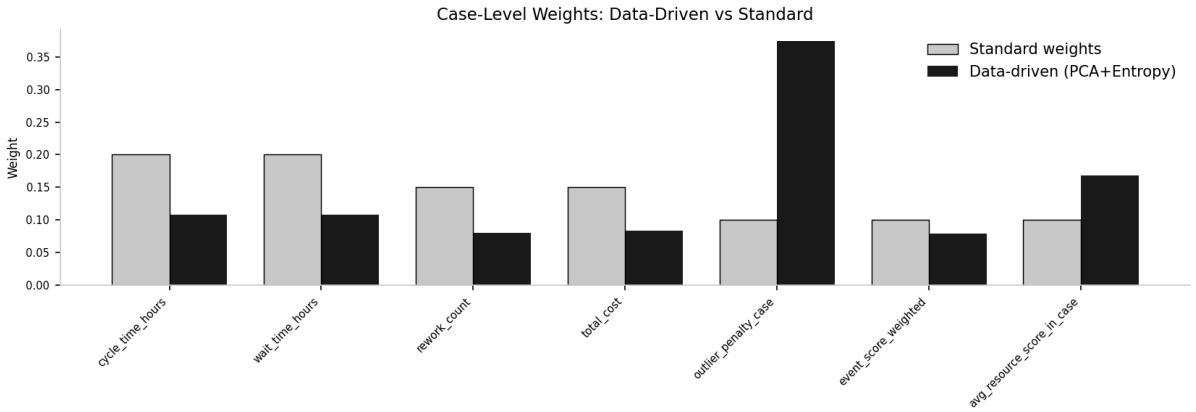

Figure 4. Comparison of standard and data-driven case-level weights

This chart contrasts manually assigned weights with PCA+entropy-derived weights. Data-driven weighting increases importance for features with higher variance or information contribution especially the outlier penalty and resource-related metrics resulting in more behaviorally meaningful scoring.

Resource-Level Results

The resource performance dashboard (Figure 5) reveals significant heterogeneity in workload distribution and behavioral characteristics. A small set of resources exhibits extremely high event volumes, handover counts, and activity diversity, while the majority operate within more moderate ranges. This inequality is typical of public-sector processes where specialized roles accumulate complexity. The hybrid resource score curve reflects this disparity, with wide performance dispersion across the workforce.

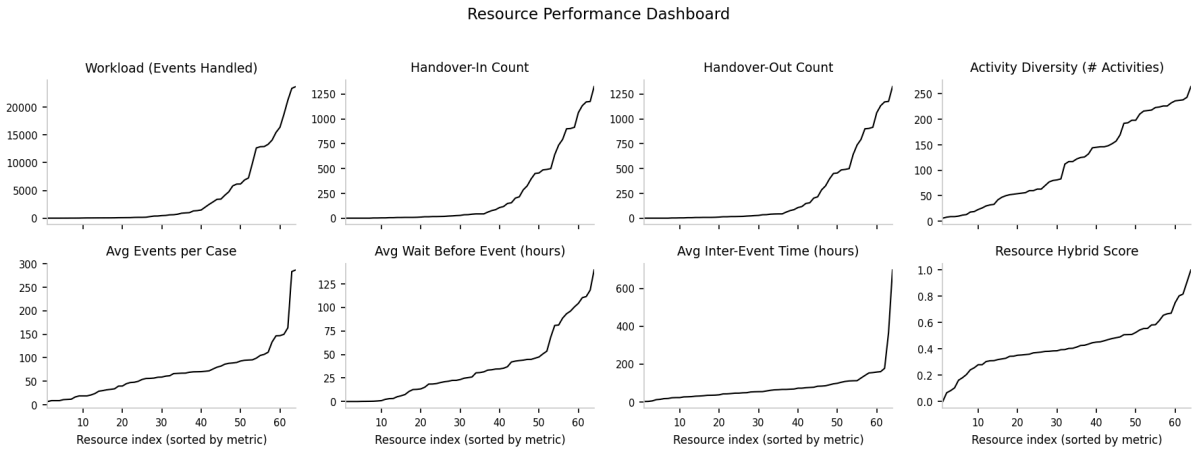

Figure 5. Resource-level performance dashboard

This multi-panel plot displays key resource behaviors, including workload, handovers, activity diversity, waiting time, and the hybrid resource score. Results show high imbalance, with a subset of resources exhibiting disproportionately high workload and collaboration intensity.Outlier effects at the resource level mirror those observed in case behavior. Figure 6 compares hybrid resource scores with and without the outlier penalty. When the penalty is removed, the distribution narrows and masks low-performing resources. With the penalty applied, structurally irregular resource behaviors become more visible, enhancing the model's sensitivity to inefficient working patterns.

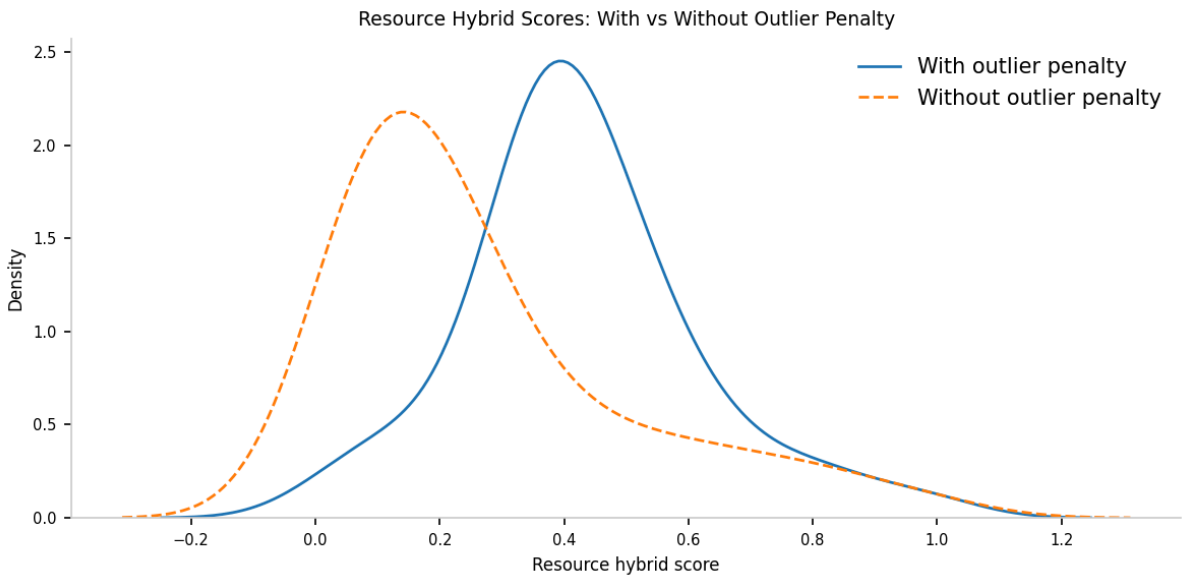

Figure 6. Distribution of resource hybrid scores with and without outlier penalties

The outlier-adjusted distribution exhibits a more distinct separation of low-performing resources, increasing the diagnostic accuracy of the performance model.

The comparative weight analysis in Figure 7 shows that data-driven weights assign higher importance to interaction- and workload-driven behaviors, such as event volume and handover frequency, which are empirically connected to performance disparities. This weighting structure ensures that the hybrid resource score emphasizes behaviors that materially influence throughput and stability.

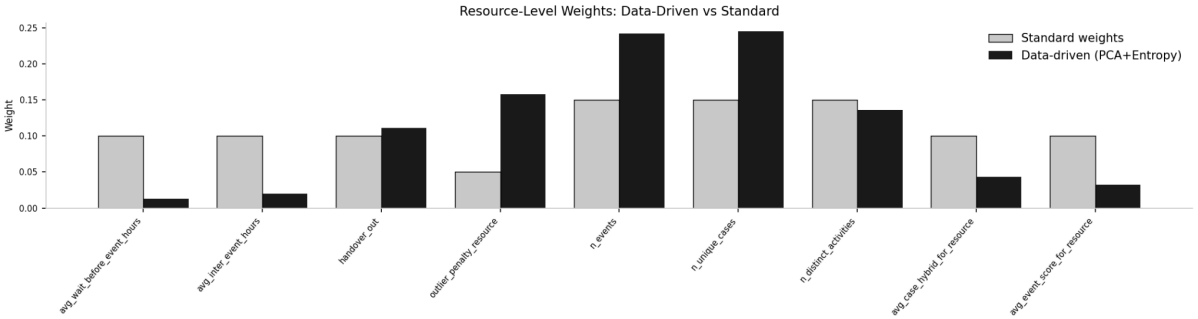

Figure 7. Comparison of standard and data-driven resource-level weights

Data-driven (PCA+entropy) weights emphasize contributions from event volume, unique cases handled, and handover interactions, reflecting their strong explanatory power in differentiating resource performance.

Municipality-Level Results

Municipality-level analysis integrates performance across cases and resources into an organizational comparison. The municipality performance dashboard (Figure 8) presents hybrid scores, average cycle times, resource processing hours, case load, and variant complexity for all municipalities. The results show clear separation between units: BPIC15_3 exhibits the strongest performance, combining moderate structural complexity with relatively low cycle times, while BPIC15_4 and BPIC15_2 show significantly weaker outcomes due to high structural complexity and long processing times.

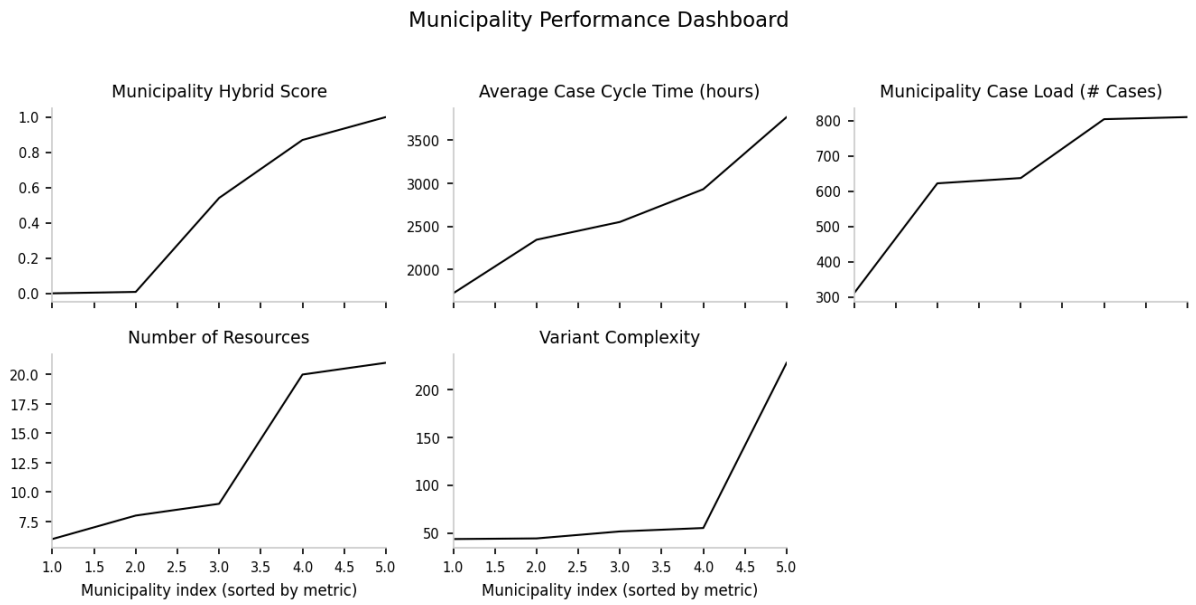

Figure 8. Municipality-level performance dashboard

Key municipality metrics hybrid score, average case cycle time, case load, resource count, and process variant complexity are presented to compare organizational performance structures and highlight inter-municipality differences.

	municipality	case_count	avg_case_cycle_hours	avg_resource_processing_hours	structural_complexity	municipality_hybrid_score	rank
2	BPIC15_3	804	1724.267544	64.611105	43.584577	1.000	1
0	BPIC15_1	810	2346.385508	106.129609	44.241975	0.870	2
4	BPIC15_5	637	2552.882832	54.468851	51.558870	0.541	3
1	BPIC15_2	310	3770.189789	65.873137	55.122581	0.008	4
3	BPIC15_4	622	2932.718776	81.202519	228.004823	0.000	5

Table 3. Municipality performance summary and hybrid score ranking

The table reports case volume, average cycle time, resource processing time, structural complexity, hybrid score, and rank for each municipality. These aggregated indicators allow comparison of organizational behavior and highlight structural inefficiencies.

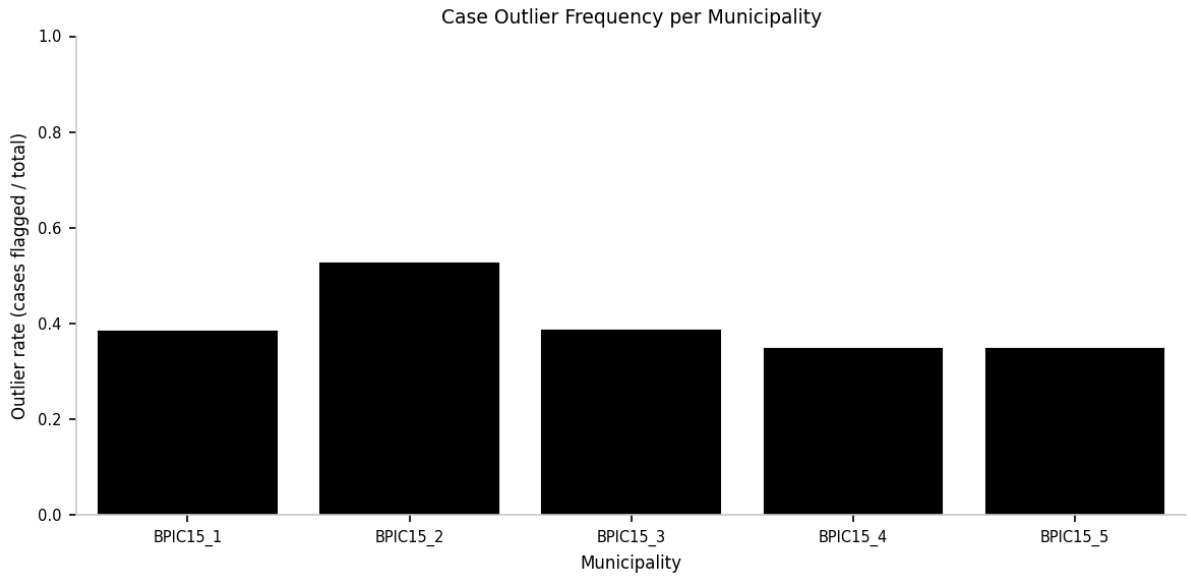

Figure 9. Case outlier frequency across municipalities

This bar chart shows the proportion of cases flagged as outliers in each municipality. Higher outlier rates indicate the presence of unstable or irregular case behaviors, revealing structural inefficiencies at the organizational level.

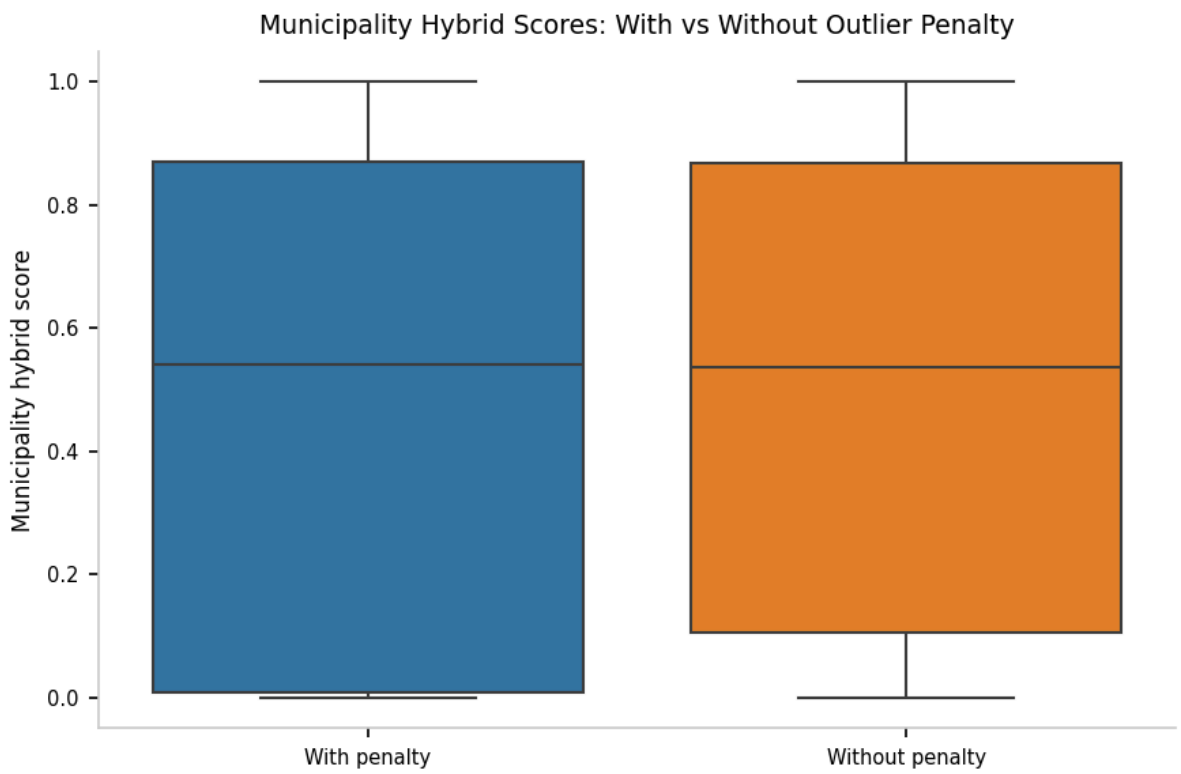

Figure 10. Municipality-level hybrid scores with and without outlier penalties

Boxplots show that outlier adjustments primarily influence lower-scoring municipalities, clarifying structural differences and improving ranking interpretability.

The distribution of outlier cases by municipality, shown in **Figure 9**, reinforces these findings. Higher outlier incidence corresponds strongly to lower hybrid scores, indicating that structural irregularities at the micro-level aggregate into reduced organizational performance. **Figure 10**, comparing municipality hybrid scores with and without penalties, demonstrates that outlier inclusion improves interpretability by increasing the separation between high- and low-performing municipalities.

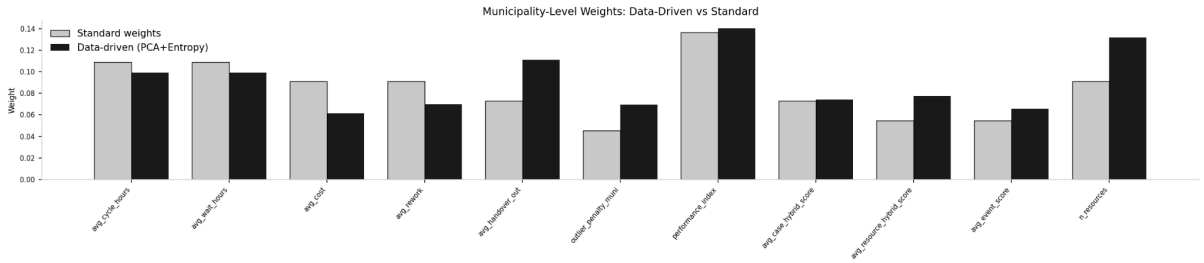

Figure 11. Standard vs data-driven municipality-level weights

The data-driven approach assigns higher weight to structural and resource-related indicators, reflecting their stronger contribution to explaining performance variation between municipalities.

Weight comparisons at the municipality level are shown in **Figure 11**, where data-driven weighting emphasizes structural and resource-based metrics. This aligns with observed performance drivers: municipalities differ most strongly in their structural complexity and resource coordination patterns, making these features more predictive of performance than evenly weighted or manually assigned schemes would suggest.

Weighting Effects and Multi-Level Score Interpretation

Across all levels of analysis case, resource, and municipality the contrast between standard and data-driven weighting schemes reveals substantial structural differences in feature importance. The PCA+entropy weighting scheme elevates the weight of features that explain the largest share of behavioral variance, leading to more accurate and meaningful ranking outputs. This effect is visible not only in the weight comparison figures (Figures 4 , 7 and 11) but also in the distributional spread of hybrid scores across levels. Figure 12 consolidates these distributions in a single comparison, showing that case-level scores are the most variable, resource scores show moderate dispersion, and municipality scores are tightly clustered due to the small sample size. This reflects the nested nature of performance variability: instance-level patterns dominate, resource behaviors moderate the effects, and organizational outcomes aggregate these influences.

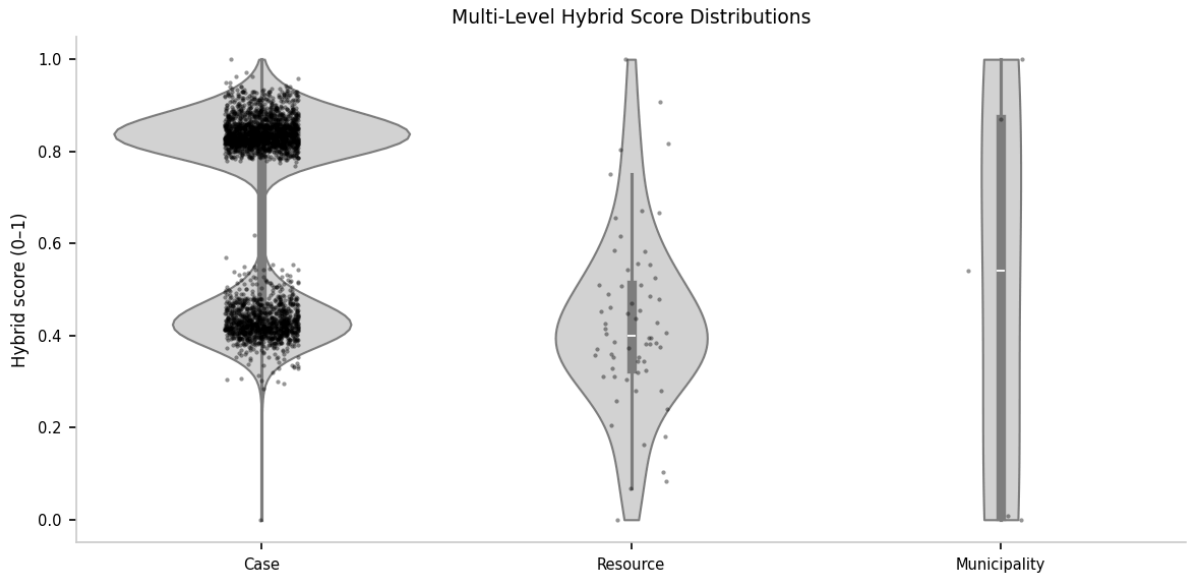

Figure 12. Comparison of hybrid score distributions across cases, resources, and municipalities

Violin plots illustrate the distribution and dispersion of hybrid scores at each layer of analysis. Cases exhibit the widest spread, resources show moderate variance, and municipalities cluster tightly due to their small sample size.

Dashboard

The dashboards presented throughout the results are not only visualization devices but functional diagnostic tools that enable deep process understanding. The case dashboard reveals structural spikes, long-cycle-time instances, and anomalous rework patterns. The resource dashboard highlights workload concentration, collaboration bottlenecks, and behavioral irregularities. The municipality dashboard contextualizes these insights at the organizational level, summarizing how micro-level behaviors aggregate into macro-level outcomes. In operational use, these dashboards enable managers to move beyond static KPIs by visually tracing where performance deteriorates and by identifying which cases, resources, or structural elements contribute most to inefficiency. The dashboard also generates a pdf format file of the report.

Overall Findings and Final Outcome

The results collectively show that the hybrid scoring system provides a far more nuanced and reliable understanding of performance compared to traditional log-based metrics or manually weighted scoring schemes. The graph-preserving reconstruction ensures that relationships between cases, events, and resources are maintained, allowing accurate computation of structural and behavioral metrics.

The multi-layer scoring reveals how irregular instance behavior propagates upward to affect resource load and eventually municipality-level performance, demonstrating interconnectedness across organizational layers. Data-driven weighting improves interpretability and ensures that metrics with real behavioral influence are prioritized. Outlier penalties ensure that structurally irregular behaviors do not disappear into aggregated averages. Finally, the dashboards synthesize these insights into accessible, actionable visual summaries that reveal where performance improvement efforts should be directed.

Conclusion

This project demonstrates a complete, data-driven approach for measuring workplace and process performance using multi-level behavioral analytics derived from the Event Graph version of the BPIC 2015 dataset. By reconstructing the dataset from its GraphML representation into a graph-preserving CSV structure, we established a reliable foundation for computing rich behavioral and structural features across cases, resources, and municipalities. This reconstruction ensured that essential graph semantics such as event ordering, entity correlations, and resource associations were retained, enabling a level of analytical precision not typically achievable in traditional event-log formats.

Building on this foundation, the project introduced a comprehensive performance measurement framework that integrates feature engineering, hybrid outlier detection, data-driven weighting, and multi-level scoring. The results reveal that workflow behavior in the municipal permit process is highly heterogeneous at the case level, unevenly distributed at the resource level, and structurally fragmented across municipalities. Outlier-aware scoring proved essential for differentiating between typical and irregular behaviors, and the comparison between standard and data-driven weight allocation demonstrated that PCA+entropy weighting more accurately reflects behavioral variance. This produces hybrid scores that are more meaningful, more discriminative, and more representative of the underlying workflow dynamics than manually weighted metrics.

The dashboards created for cases, resources, and municipalities acted as operational diagnostic tools, synthesizing complex behavioral indicators into clear, interpretable visual summaries. These dashboards revealed extreme case durations, workload imbalances, collaboration bottlenecks, structural spikes, and process irregularities. Moreover, the combination of dashboards, weight comparisons, and outlier-adjusted scoring highlighted the interconnected nature of performance across organizational layers: irregular case behavior propagates to resource pressure, which in turn aggregates into municipal performance disparities. This multi-layer insight is directly aligned with the goals of the Workplace Performance Measurement (WPM) approach, but achieved here through algorithmic reconstruction rather than sensor-based observation, demonstrating a novel computational pathway for high-resolution performance analysis.

The project's contributions extend beyond descriptive analytics. It closes key research gaps in the WPM and Event Graph literature by demonstrating how data-driven hybrid scoring can be integrated with graph-derived behavioral features to support performance evaluation across multiple organizational layers. Unlike traditional process mining, which typically emphasizes control-flow discovery and bottleneck detection, this work emphasizes behavioral regularity, structural complexity, anomaly influence, and cross-layer interpretability.

Unlike the event-graph research, which focuses on data representation, this work develops a full analytical pipeline and scoring methodology on top of that representation. The result is a robust framework suitable for operational decision-making, continuous improvement, and organizational benchmarking.

Overall, the system developed in this project provides a scalable, reliable, and behaviorally grounded approach to performance measurement. By integrating graph semantics, statistical rigor, machine learning, and multi-level analytics, it offers a comprehensive view of workflow performance that can guide targeted interventions, resource allocation, process redesign, and cross-municipality benchmarking. Future work may extend this framework by incorporating temporal drift analysis, stability testing, predictive modeling, or direct integration with WPM's sensor-based observation methods to create a unified analytical–physical measurement ecosystem. Nonetheless, the present results already demonstrate the substantial value of a data-driven, hybrid, graph-enhanced approach to workplace performance measurement.

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